

Asset Management System

Forms, Database Structure and Functional Explanation

1. System Architecture Overview

The Asset Management System should be developed as a centralized web application containing the following layers:

1. User Interface
2. Application Programming Interface
3. Business Logic
4. Database
5. Document Storage
6. Notification Service
7. Reporting Service
8. Audit and Security Layer

A recommended technology stack is:

Frontend

- React
- TypeScript
- Material UI or Bootstrap
- React Hook Form
- Data tables with filtering and export

Backend

- Node.js with NestJS
or
- ASP.NET Core Web API

Database

- PostgreSQL

File Storage

- Local secure file server
- Cloud storage
- Amazon S3-compatible storage
- Azure Blob Storage

The actual files should be stored in secure storage. The database should store:

- File name
 - File path
 - Document type
 - Related asset
 - Issue date
 - Expiry date
 - Version
 - Uploaded user
 - Uploaded date
-

2. Main System Modules

The Asset Management System should include:

1. Dashboard
 2. General Asset Register
 3. Vehicle and Fleet Management
 4. Toll Management
 5. Maintenance Management
 6. Registration and Insurance
 7. Permit Management
 8. Fine Management
 9. Accident Management
 10. Telecom and SIM Management
 11. Document Management
 12. Expense Management
 13. Alerts and Notifications
 14. Workflow and Approvals
 15. Reports
 16. User and Permission Management
 17. System Settings
 18. Audit Trail
-

3. General Form Design Standards

All forms should follow the same design structure.

3.1 Form Header

The header should display:

- Form title
- Record number
- Current status
- Company
- Branch
- Created date
- Created by
- Last modified date
- Last modified by

3.2 Standard Form Buttons

Each form should provide appropriate buttons:

- Save Draft
- Save and Submit
- Update
- Cancel
- Delete
- Approve
- Reject
- Print
- Download
- Upload Document
- View History
- Close Record

3.3 Standard Form Validations

The system should validate:

- Mandatory fields
- Duplicate records
- Date ranges
- Expiry dates
- Numeric values
- Negative amounts
- Invalid odometer readings
- Invalid mobile numbers
- Duplicate chassis numbers
- Duplicate SIM serial numbers
- Duplicate invoice references
- Unsupported file types

- Maximum upload size

3.4 Standard Record Statuses

Common statuses should include:

- Draft
 - Submitted
 - Pending Approval
 - Approved
 - Rejected
 - Active
 - Inactive
 - Completed
 - Cancelled
 - Expired
 - Closed
-

4. Company Form

4.1 Purpose

The Company Form is used to create and maintain the legal entities or business units that own or manage assets.

4.2 Form Fields

Company Information

- Company Code
- Company Name
- Trade Name
- Tax Registration Number
- Licence Number
- Licence Expiry Date
- Phone
- Email
- Website
- Address
- Emirate
- Country
- Logo
- Active Status

4.3 Database Table

companies

Column	Type	Description
id	BIGINT	Primary key
company_code	VARCHAR(30)	Unique company code
company_name	VARCHAR(150)	Legal company name
trade_name	VARCHAR(150)	Commercial name
tax_registration_number	VARCHAR(50)	VAT registration number
licence_number	VARCHAR(50)	Trade licence number
licence_expiry_date	DATE	Licence expiry
phone	VARCHAR(30)	Main contact
email	VARCHAR(150)	Main email
website	VARCHAR(200)	Company website
address	TEXT	Company address
emirate_id	BIGINT	Related emirate
country_id	BIGINT	Related country
logo_path	VARCHAR(500)	Logo storage path
is_active	BOOLEAN	Active status
created_at	TIMESTAMP	Creation date
created_by	BIGINT	Creating user
updated_at	TIMESTAMP	Modified date
updated_by	BIGINT	Modifying user

4.4 Validation

- Company Code is mandatory and unique.
 - Company Name is mandatory.
 - Email must use a valid email format.
 - Licence expiry date cannot be earlier than the licence issue date.
-

5. Branch Form

5.1 Purpose

The Branch Form manages offices, warehouses, shops, operating sites, and other physical locations.

5.2 Form Fields

- Company
- Branch Code
- Branch Name
- Branch Type
- Emirate
- Address
- Contact Person
- Contact Number
- Email
- Cost Centre
- Active Status

5.3 Database Table

branches

Column	Type	Description
id	BIGINT	Primary key
company_id	BIGINT	Foreign key to companies
branch_code	VARCHAR(30)	Branch code
branch_name	VARCHAR(150)	Branch name
branch_type	VARCHAR(50)	Office, warehouse or shop
emirate_id	BIGINT	Related emirate
address	TEXT	Branch address
contact_person	VARCHAR(150)	Branch contact
contact_number	VARCHAR(30)	Contact number
email	VARCHAR(150)	Branch email
cost_center_id	BIGINT	Related cost centre
is_active	BOOLEAN	Active status

6. Department Form

6.1 Form Fields

- Company
- Branch
- Department Code
- Department Name
- Department Manager
- Cost Centre
- Active Status
- Remarks

6.2 Database Table

departments

Column	Type	Description
id	BIGINT	Primary key
company_id	BIGINT	Company
branch_id	BIGINT	Branch
department_code	VARCHAR(30)	Unique code
department_name	VARCHAR(150)	Department name
manager_employee_id	BIGINT	Department manager
cost_center_id	BIGINT	Cost centre
is_active	BOOLEAN	Active status
remarks	TEXT	Remarks

7. Employee Form

7.1 Purpose

The Employee Form manages employees who may receive vehicles, SIM cards, phones, laptops, access cards, or other assets.

7.2 Form Fields

Basic Information

- Employee Code
- Full Name
- Company
- Branch
- Department
- Job Title
- Joining Date
- Employment Status
- Resignation Date

Contact Information

- Work Email
- Personal Email
- Work Mobile
- Personal Mobile

Identification

- Emirates ID Number
- Emirates ID Expiry Date
- Passport Number
- Passport Expiry Date
- Driving Licence Number
- Driving Licence Expiry Date
- Driving Licence Category

Driver Information

- Is Driver
- Driver Type
- Allowed Vehicle Categories
- Driver Status

Documents

- Emirates ID Copy
- Passport Copy
- Driving Licence Copy
- Employee Photograph

7.3 Database Table

employees

Column	Type	Description
id	BIGINT	Primary key
employee_code	VARCHAR(30)	Unique employee number
company_id	BIGINT	Company
branch_id	BIGINT	Branch
department_id	BIGINT	Department
full_name	VARCHAR(150)	Employee name
job_title	VARCHAR(100)	Job title
joining_date	DATE	Joining date
employment_status	VARCHAR(30)	Active, resigned or terminated
resignation_date	DATE	Last working date
work_email	VARCHAR(150)	Official email
personal_email	VARCHAR(150)	Personal email
work_mobile	VARCHAR(30)	Official number
personal_mobile	VARCHAR(30)	Personal number
emirates_id_number	VARCHAR(30)	Restricted identification
emirates_id_expiry	DATE	Expiry date
passport_number	VARCHAR(30)	Passport number
passport_expiry	DATE	Passport expiry
driving_licence_number	VARCHAR(50)	Licence number
driving_licence_expiry	DATE	Licence expiry
is_driver	BOOLEAN	Driver indicator
driver_type	VARCHAR(50)	Light or heavy driver
is_active	BOOLEAN	Active status

7.4 Validation

- Employee Code must be unique.

- Full Name is mandatory.
 - Resignation date is required when status is resigned.
 - Driving licence details are required when Is Driver is selected.
 - Licence expiry alert should be generated automatically.
-

8. Asset Category Form

8.1 Form Fields

- Category Code
- Category Name
- Parent Category
- Description
- Asset Number Prefix
- Depreciable Asset
- Active Status

8.2 Database Table

asset_categories

Column	Type	Description
id	BIGINT	Primary key
category_code	VARCHAR(30)	Unique code
category_name	VARCHAR(100)	Category name
parent_category_id	BIGINT	Parent category
description	TEXT	Category description
asset_number_prefix	VARCHAR(20)	Asset code prefix
is_depreciable	BOOLEAN	Depreciation applicability
is_active	BOOLEAN	Active status

Example categories:

- Vehicle
- Telecom
- SIM Card
- Mobile Device
- Laptop
- Furniture
- Machinery

- Office Equipment
-

9. General Asset Registration Form

9.1 Purpose

This form registers every asset in the centralized asset register.

9.2 Form Fields

Identification

- Asset Code
- Asset Name
- Asset Category
- Asset Type
- Serial Number
- Barcode
- QR Code

Ownership

- Company
- Branch
- Department
- Ownership Type
- Supplier
- Purchase Date
- Purchase Cost
- Invoice Number

Asset Details

- Brand
- Model
- Description
- Current Location
- Condition
- Current Status

Assignment

- Assigned Employee
- Assignment Date
- Expected Return Date

Financial Information

- Cost Centre
- Depreciation Method
- Useful Life
- Residual Value

Documents

- Purchase Invoice
- Warranty
- Asset Photograph
- Handover Form
- Other Attachments

9.3 Database Table

assets

Column	Type	Description
id	BIGINT	Primary key
asset_code	VARCHAR(50)	Unique asset number
asset_category_id	BIGINT	Asset category
asset_type_id	BIGINT	Asset type
company_id	BIGINT	Owning company
branch_id	BIGINT	Branch
department_id	BIGINT	Department
asset_name	VARCHAR(150)	Asset name
brand	VARCHAR(100)	Brand
model	VARCHAR(100)	Model
serial_number	VARCHAR(100)	Serial number
barcode	VARCHAR(100)	Barcode value
qr_code	VARCHAR(200)	QR code value
description	TEXT	Description
ownership_type_id	BIGINT	Ownership type
supplier_id	BIGINT	Supplier
acquisition_date	DATE	Purchase date

Column	Type	Description
acquisition_cost	DECIMAL(18,2)	Purchase cost
invoice_number	VARCHAR(100)	Purchase invoice
cost_center_id	BIGINT	Cost centre
asset_status_id	BIGINT	Current status
current_employee_id	BIGINT	Current holder
current_location	VARCHAR(250)	Current location
condition_status	VARCHAR(50)	Asset condition
is_active	BOOLEAN	Active status

10. Asset Assignment Form

10.1 Purpose

The form records the issuing, transferring, and returning of assets.

10.2 Form Fields

- Assignment Number
- Asset
- Employee
- Company
- Branch
- Department
- Assignment Date
- Expected Return Date
- Asset Condition at Handover
- Accessories Issued
- Employee Acknowledgement
- Handover Document
- Assigned By
- Remarks

10.3 Return Section

- Return Date
- Asset Condition at Return
- Missing Accessories
- Damage Description

- Return Accepted By
- Recovery Amount
- Return Remarks

10.4 Database Table

asset_assignments

Column	Type	Description
id	BIGINT	Primary key
assignment_number	VARCHAR(50)	Unique assignment number
asset_id	BIGINT	Assigned asset
employee_id	BIGINT	Assigned employee
company_id	BIGINT	Company
branch_id	BIGINT	Branch
department_id	BIGINT	Department
assignment_date	DATE	Issue date
expected_return_date	DATE	Planned return date
actual_return_date	DATE	Actual return date
condition_at_handover	VARCHAR(100)	Handover condition
condition_at_return	VARCHAR(100)	Return condition
accessories_issued	TEXT	Accessories
missing_accessories	TEXT	Missing items
damage_description	TEXT	Return damage
recovery_amount	DECIMAL(18,2)	Employee recovery
assignment_status	VARCHAR(30)	Assigned, returned or transferred
assigned_by	BIGINT	Issuing user
returned_to	BIGINT	Receiving user
acknowledgement_document_id	BIGINT	Handover document
remarks	TEXT	Remarks

10.5 Business Rules

- Only one active assignment should exist per asset.

- An inactive employee cannot receive an asset.
 - An asset under maintenance cannot be assigned.
 - Returning an asset should update the asset status to unassigned.
 - Transferring an asset should close the previous assignment.
-

11. Vehicle Registration Form

11.1 Form Fields

Asset Information

- Asset Code
- Company
- Branch
- Department
- Vehicle Category

Vehicle Identification

- Plate Number
- Plate Emirate
- Plate Category
- Chassis Number
- Engine Number
- Registration Number

Vehicle Details

- Make
- Model
- Manufacturing Year
- Colour
- Fuel Type
- Transmission
- Seating Capacity
- Gross Vehicle Weight
- Chiller Available
- Trailer Type

Purchase Information

- Ownership Type
- Supplier
- Purchase Date
- Purchase Price
- Finance Company

- Lease Start Date
- Lease End Date

Assignment

- Assigned Driver
- Assigned Department
- Current Location
- Vehicle Status

Documents

- Registration Card
- Purchase Invoice
- Vehicle Photograph
- Ownership Document
- Finance Agreement

11.2 Database Tables

vehicles

Column	Type	Description
id	BIGINT	Primary key
asset_id	BIGINT	Unique foreign key to assets
vehicle_category_id	BIGINT	Vehicle category
vehicle_make_id	BIGINT	Make
vehicle_model_id	BIGINT	Model
plate_number	VARCHAR(30)	Vehicle plate
plate_emirate_id	BIGINT	Plate emirate
plate_category	VARCHAR(30)	Plate type
registration_number	VARCHAR(50)	Registration reference
chassis_number	VARCHAR(100)	Unique chassis number
engine_number	VARCHAR(100)	Engine number
manufacturing_year	INTEGER	Model year
color	VARCHAR(50)	Vehicle colour
fuel_type	VARCHAR(30)	Fuel
transmission_type	VARCHAR(30)	Transmission

Column	Type	Description
seating_capacity	INTEGER	Seats
gross_vehicle_weight	DECIMAL(12,2)	Vehicle weight
current_odometer	DECIMAL(14,2)	Latest verified mileage
has_chiller	BOOLEAN	Chiller indicator
assigned_driver_id	BIGINT	Assigned driver
vehicle_status	VARCHAR(30)	Current status

vehicle_categories

- id
- category_code
- category_name
- parent_category_id
- is_active

vehicle_makes

- id
- make_name
- is_active

vehicle_models

- id
- vehicle_make_id
- model_name
- is_active

11.3 Validation

- Chassis Number must be unique.
- Plate Number with Plate Emirate must be unique.
- Manufacturing Year should not be greater than the current year plus one.
- Assigned Driver must have a valid driving licence.
- Heavy vehicles can only be assigned to authorized heavy-vehicle drivers.

12. Vehicle Registration Renewal Form

12.1 Form Fields

- Renewal Number
- Vehicle
- Current Registration Expiry
- Passing Test Date
- Passing Test Result
- New Registration Start Date
- New Registration Expiry Date
- Insurance Provider
- Insurance Policy
- Testing Fee
- Registration Fee
- Insurance Amount
- Other Charges
- VAT
- Total Amount
- Renewal Status
- Responsible Employee
- Payment Reference
- Attachments

12.2 Database Table

vehicle_registrations

Column	Type	Description
id	BIGINT	Primary key
vehicle_id	BIGINT	Related vehicle
registration_number	VARCHAR(50)	Registration number
issue_date	DATE	Issue date
expiry_date	DATE	Expiry date
passing_test_date	DATE	Passing date
passing_result	VARCHAR(30)	Passed or failed
renewal_status	VARCHAR(30)	Pending or completed
renewal_completed_date	DATE	Completion date
testing_fee	DECIMAL(18,2)	Passing charge

Column	Type	Description
renewal_fee	DECIMAL(18,2)	Registration charge
insurance_amount	DECIMAL(18,2)	Insurance cost
other_charges	DECIMAL(18,2)	Other costs
tax_amount	DECIMAL(18,2)	VAT
total_amount	DECIMAL(18,2)	Total
responsible_employee_id	BIGINT	Responsible person
document_id	BIGINT	Registration document

12.3 Explanation

A new registration record should be created for each renewal. Previous renewal records should not be overwritten. This provides a complete registration history.

13. Vehicle Insurance Form

13.1 Form Fields

- Vehicle
- Insurance Provider
- Policy Number
- Policy Type
- Coverage Type
- Start Date
- Expiry Date
- Premium Amount
- Excess Amount
- Covered Drivers
- Geographic Coverage
- Roadside Assistance
- Policy Status
- Insurance Document
- Payment Receipt
- Remarks

13.2 Database Table

vehicle_insurance_policies

Column	Type	Description
id	BIGINT	Primary key
vehicle_id	BIGINT	Related vehicle
insurance_provider_id	BIGINT	Supplier or provider
policy_number	VARCHAR(100)	Policy number
policy_type	VARCHAR(50)	Policy type
coverage_type	VARCHAR(50)	Comprehensive or third party
start_date	DATE	Start date
expiry_date	DATE	Expiry date
premium_amount	DECIMAL(18,2)	Premium
excess_amount	DECIMAL(18,2)	Excess
geographic_coverage	TEXT	Coverage area
roadside_assistance	BOOLEAN	Included or not
policy_status	VARCHAR(30)	Active or expired
document_id	BIGINT	Policy document

14. Toll Account Form

14.1 Form Fields

- Toll Provider
- Account Name
- Account Number
- Account Owner Type
- Company
- Account Holder
- Current Balance
- Minimum Balance
- Recharge Method
- Account Status
- Remarks

14.2 Database Table

toll_accounts

Column	Type	Description
id	BIGINT	Primary key
provider_id	BIGINT	Salik or Darb
company_id	BIGINT	Company
account_name	VARCHAR(150)	Account name
account_number	VARCHAR(100)	Account number
ownership_type	VARCHAR(30)	Corporate or personal
account_holder_employee_id	BIGINT	Personal account holder
current_balance	DECIMAL(18,2)	Current balance
minimum_balance	DECIMAL(18,2)	Alert threshold
recharge_method	VARCHAR(50)	Recharge method
account_status	VARCHAR(30)	Account status
remarks	TEXT	Remarks

15. Toll Tag Assignment Form

15.1 Form Fields

- Toll Provider
- Toll Account
- Tag Number
- Vehicle
- Assignment Date
- Removal Date
- Opening Balance
- Tag Status
- Remarks

15.2 Database Table

toll_tags

Column	Type	Description
id	BIGINT	Primary key
toll_account_id	BIGINT	Toll account
tag_number	VARCHAR(100)	Unique tag
vehicle_id	BIGINT	Current vehicle
assignment_date	DATE	Assigned date
removal_date	DATE	Removed date
status	VARCHAR(30)	Active, removed or blocked
remarks	TEXT	Remarks

A separate history table is recommended:

toll_tag_assignment_history

- id
- toll_tag_id
- vehicle_id
- assigned_date
- removed_date
- assigned_by
- remarks

16. Toll Transaction Import Form

16.1 Form Fields

- Toll Provider
- Toll Account
- Import Month
- File Upload
- File Type
- Number of Records
- Duplicate Records
- Invalid Records
- Unmatched Vehicles

- Import Status

16.2 Preview Grid

The preview should display:

- Transaction Date
- Transaction Time
- Toll Gate
- Tag Number
- Plate Number
- Vehicle
- Amount
- Duplicate Status
- Validation Message

16.3 Database Tables

toll_import_batches

- id
- toll_account_id
- import_month
- file_name
- file_path
- total_records
- valid_records
- invalid_records
- duplicate_records
- import_status
- imported_by
- imported_at

toll_transactions

Column	Type	Description
id	BIGINT	Primary key
import_batch_id	BIGINT	Related import
toll_account_id	BIGINT	Toll account
toll_tag_id	BIGINT	Toll tag
vehicle_id	BIGINT	Vehicle
transaction_date	DATE	Date

Column	Type	Description
transaction_time	TIME	Time
toll_gate	VARCHAR(150)	Gate name
plate_number	VARCHAR(30)	Reported plate
transaction_reference	VARCHAR(100)	External reference
amount	DECIMAL(18,2)	Toll amount
matching_status	VARCHAR(30)	Matched or unmatched
is_duplicate	BOOLEAN	Duplicate indicator

17. Odometer Update Form

17.1 Form Fields

- Vehicle
- Reading Date
- Odometer Reading
- Reading Source
- Driver
- Dashboard Photograph
- Submitted By
- Verification Status
- Verified By
- Remarks

17.2 Database Table

vehicle_odometer_readings

Column	Type	Description
id	BIGINT	Primary key
vehicle_id	BIGINT	Vehicle
reading_date	DATE	Reading date
odometer_reading	DECIMAL(14,2)	Mileage
reading_source	VARCHAR(30)	Manual, photo or service
driver_id	BIGINT	Driver

Column	Type	Description
photo_document_id	BIGINT	Dashboard photograph
verification_status	VARCHAR(30)	Pending, verified or rejected
verified_by	BIGINT	Verifying user
verified_at	TIMESTAMP	Verification time
remarks	TEXT	Remarks

17.3 Validation

- New reading cannot normally be less than the previous verified reading.
 - Large mileage differences should require confirmation.
 - Only verified readings should update the vehicle's current odometer.
 - Missing weekly readings should generate alerts.
-

18. Service Contract Form

18.1 Form Fields

- Vehicle
- Service Provider
- Contract Number
- Start Date
- Expiry Date
- Starting Odometer
- Maximum Covered Mileage
- Number of Included Services
- Minor Services Included
- Major Services Included
- Contract Value
- Included Services
- Excluded Services
- Contact Person
- Contract Document
- Status
- Remarks

18.2 Database Table

vehicle_service_contracts

Column	Type	Description
id	BIGINT	Primary key
vehicle_id	BIGINT	Vehicle
service_provider_id	BIGINT	Supplier
contract_number	VARCHAR(100)	Contract number
start_date	DATE	Start date
expiry_date	DATE	Expiry date
starting_odometer	DECIMAL(14,2)	Starting mileage
mileage_limit	DECIMAL(14,2)	Maximum mileage
included_service_count	INTEGER	Included visits
contract_value	DECIMAL(18,2)	Contract cost
included_services	TEXT	Covered services
excluded_services	TEXT	Exclusions
status	VARCHAR(30)	Active or expired
document_id	BIGINT	Contract file

19. Service Schedule Form

19.1 Form Fields

- Vehicle
- Service Type
- Service Name
- Interval Kilometres
- Interval Months
- Last Service Date
- Last Service Odometer
- Next Service Date
- Next Service Odometer
- Alert Before Kilometres
- Alert Before Days

- Active Status

19.2 Database Table

vehicle_service_schedules

Column	Type	Description
id	BIGINT	Primary key
vehicle_id	BIGINT	Vehicle
service_type_id	BIGINT	Minor, major or custom
service_name	VARCHAR(150)	Service name
interval_kilometres	DECIMAL(14,2)	Mileage interval
interval_months	INTEGER	Time interval
last_service_date	DATE	Previous service date
last_service_odometer	DECIMAL(14,2)	Previous mileage
next_service_date	DATE	Calculated due date
next_service_odometer	DECIMAL(14,2)	Calculated due mileage
alert_before_kilometres	DECIMAL(14,2)	Mileage threshold
alert_before_days	INTEGER	Date threshold
is_active	BOOLEAN	Active status

19.3 Calculation Rule

The service is due when either condition occurs first:

- Current odometer reaches the next service odometer.
- Current date reaches the next service date.

20. Maintenance Request Form

20.1 Form Fields

- Request Number
- Vehicle
- Request Date
- Reported By

- Driver
- Current Odometer
- Issue Category
- Issue Description
- Priority
- Vehicle Operational Status
- Photograph
- Preferred Workshop
- Estimated Cost
- Approval Status

20.2 Database Table

maintenance_requests

Column	Type	Description
id	BIGINT	Primary key
request_number	VARCHAR(50)	Unique request
vehicle_id	BIGINT	Vehicle
request_date	DATE	Request date
reported_by	BIGINT	Employee
driver_id	BIGINT	Driver
odometer_reading	DECIMAL(14,2)	Mileage
issue_category_id	BIGINT	Issue type
issue_description	TEXT	Problem
priority	VARCHAR(20)	Low, medium, high or urgent
operational_status	VARCHAR(30)	Running or stopped
preferred_supplier_id	BIGINT	Workshop
estimated_cost	DECIMAL(18,2)	Estimate
approval_status	VARCHAR(30)	Approval state
request_status	VARCHAR(30)	Open, approved or closed

21. Maintenance Job Form

21.1 Form Fields

Job Information

- Job Number
- Maintenance Request
- Vehicle
- Service Type
- Workshop
- Service Date
- Odometer
- Complaint
- Inspection Findings
- Work Performed

Cost Information

- Labour Cost
- Parts Cost
- Other Cost
- Discount
- VAT
- Total Cost

Completion Information

- Vehicle Received Date
- Vehicle Returned Date
- Downtime
- Warranty Applicable
- Next Service Date
- Next Service Odometer
- Job Status

Attachments

- Estimate
- Approval
- Job Card
- Invoice
- Before Photograph
- After Photograph

21.2 Database Tables

maintenance_jobs

Column	Type	Description
id	BIGINT	Primary key
job_number	VARCHAR(50)	Unique job
maintenance_request_id	BIGINT	Related request
vehicle_id	BIGINT	Vehicle
service_type_id	BIGINT	Service type
supplier_id	BIGINT	Workshop
service_date	DATE	Service date
odometer_reading	DECIMAL(14,2)	Mileage
complaint	TEXT	Reported problem
inspection_findings	TEXT	Findings
work_performed	TEXT	Completed work
labour_cost	DECIMAL(18,2)	Labour
parts_cost	DECIMAL(18,2)	Parts
other_cost	DECIMAL(18,2)	Other
discount_amount	DECIMAL(18,2)	Discount
tax_amount	DECIMAL(18,2)	VAT
total_cost	DECIMAL(18,2)	Total
received_date	DATE	Workshop receipt
returned_date	DATE	Vehicle return
downtime_days	INTEGER	Downtime
warranty_applicable	BOOLEAN	Warranty status
job_status	VARCHAR(30)	Open or completed

maintenance_job_parts

- id
- maintenance_job_id
- part_id

- part_description
 - quantity
 - unit_cost
 - total_cost
 - warranty_start_date
 - warranty_expiry_date
 - warranty_mileage
-

22. Parts and Warranty Form

22.1 Form Fields

- Vehicle
- Part Category
- Part Name
- Brand
- Model
- Serial Number
- Supplier
- Purchase Date
- Installation Date
- Odometer at Installation
- Warranty Period
- Warranty Expiry Date
- Warranty Mileage
- Invoice
- Warranty Card
- Current Status
- Claim Status

22.2 Database Table

vehicle_parts

Column	Type	Description
id	BIGINT	Primary key
vehicle_id	BIGINT	Vehicle
part_category_id	BIGINT	Part category
part_name	VARCHAR(150)	Part name
brand	VARCHAR(100)	Brand
model	VARCHAR(100)	Model

Column	Type	Description
serial_number	VARCHAR(100)	Serial number
supplier_id	BIGINT	Supplier
purchase_date	DATE	Purchase date
installation_date	DATE	Installation date
odometer_at_installation	DECIMAL(14,2)	Installation mileage
warranty_expiry_date	DATE	Date warranty
warranty_mileage	DECIMAL(14,2)	Mileage warranty
removal_date	DATE	Removal date
odometer_at_removal	DECIMAL(14,2)	Removal mileage
status	VARCHAR(30)	Installed or removed
warranty_claim_status	VARCHAR(30)	Claim status

23. Tyre Management Form

23.1 Form Fields

- Vehicle
- Tyre ID
- Brand
- Model
- Size
- Serial Number
- Manufacturing Date
- Supplier
- Purchase Cost
- Installation Date
- Installation Odometer
- Wheel Position
- Warranty Expiry
- Warranty Mileage
- Current Condition
- Removal Date
- Removal Odometer
- Removal Reason

23.2 Database Table

vehicle_tyres

- id
- vehicle_id
- tyre_code
- brand
- model
- tyre_size
- serial_number
- manufacturing_date
- supplier_id
- purchase_cost
- installation_date
- installation_odometer
- wheel_position
- warranty_expiry_date
- warranty_mileage
- current_condition
- removal_date
- removal_odometer
- removal_reason
- status

The system should allow tyre movement between wheel positions while maintaining history.

24. Traffic Fine Form

24.1 Form Fields

- Fine Reference
- Vehicle
- Driver
- Fine Date
- Fine Time
- Emirate
- Authority
- Violation Type
- Description
- Fine Amount
- Black Points
- Impoundment Days
- Payment Responsibility
- Salary Deduction Required

- Fine Status
- Payment Date
- Receipt
- Fine Document
- Remarks

24.2 Database Table

traffic_fines

Column	Type	Description
id	BIGINT	Primary key
fine_reference	VARCHAR(100)	Unique fine reference
vehicle_id	BIGINT	Vehicle
driver_id	BIGINT	Responsible driver
fine_date	DATE	Fine date
fine_time	TIME	Fine time
emirate_id	BIGINT	Emirate
authority_name	VARCHAR(150)	Fine authority
violation_type_id	BIGINT	Violation type
violation_description	TEXT	Description
fine_amount	DECIMAL(18,2)	Fine amount
black_points	INTEGER	Black points
impoundment_days	INTEGER	Impoundment
responsibility_type	VARCHAR(30)	Company or employee
company_paid_amount	DECIMAL(18,2)	Company amount
employee_paid_amount	DECIMAL(18,2)	Employee amount
salary_deduction_status	VARCHAR(30)	Deduction state
fine_status	VARCHAR(30)	Pending or cleared
payment_date	DATE	Payment date
receipt_document_id	BIGINT	Receipt
fine_document_id	BIGINT	Fine document

25. Delivery Permit Form

25.1 Form Fields

- Vehicle
- Permit Type
- Permit Number
- Authority
- Emirate
- Issue Date
- Expiry Date
- Application Date
- Permit Cost
- Renewal Status
- Responsible Employee
- Permit Document
- Supporting Documents
- Remarks

25.2 Database Table

vehicle_permits

- id
 - vehicle_id
 - permit_type_id
 - permit_number
 - authority_name
 - emirate_id
 - application_date
 - issue_date
 - expiry_date
 - permit_cost
 - renewal_status
 - responsible_employee_id
 - document_id
 - status
 - remarks
-

26. Accident and Incident Form

26.1 Form Fields

Incident Information

- Incident Number
- Vehicle
- Driver
- Incident Type
- Incident Date
- Incident Time
- Location
- Emirate
- Description
- Damage Description

Police and Third Party

- Police Report Number
- Police Authority
- Third-Party Involved
- Third-Party Details
- Driver Fault Status

Insurance

- Insurance Claim Number
- Insurance Provider
- Claim Status
- Approved Claim Amount

Repair

- Workshop
- Repair Estimate
- Final Repair Cost
- Parts Replaced
- Repair Description
- Vehicle Downtime

Documents

- Police Report
- Scene Photographs
- Damage Photographs
- Estimate

- Invoice
- Insurance Claim
- Completion Photographs

26.2 Database Table

vehicle_incidents

- id
- incident_number
- vehicle_id
- driver_id
- incident_type_id
- incident_date
- incident_time
- location
- emirate_id
- incident_description
- damage_description
- third_party_involved
- third_party_details
- police_report_number
- driver_fault_status
- insurance_claim_number
- insurance_claim_status
- insurance_approved_amount
- supplier_id
- estimated_repair_cost
- final_repair_cost
- company_paid_amount
- employee_liability_amount
- downtime_days
- incident_status
- closure_date

27. SIM Card Registration Form

27.1 Form Fields

SIM Information

- SIM Asset Code
- Mobile Number
- SIM Serial Number
- ICCID

- PUK Number
- Telecom Provider
- Billing Account
- Account Number

Plan Information

- Subscription Plan
- Monthly Rental
- Data Allowance
- Call Allowance
- SMS Allowance
- Contract Start Date
- Contract Expiry Date

Service Controls

- Roaming Enabled
- International Calls Enabled
- Premium SMS Enabled
- Data Roaming Enabled

Assignment

- Assigned Employee
- Department
- Assignment Date
- Device
- Device IMEI

Status

- Active
- Unassigned
- Suspended
- Blocked
- Lost
- Cancelled

Documents

- SIM Agreement
- Handover Form
- Replacement Document
- Cancellation Document

27.2 Database Table

sim_cards

Column	Type	Description
id	BIGINT	Primary key
asset_id	BIGINT	Related general asset
mobile_number	VARCHAR(30)	Unique number
sim_serial_number	VARCHAR(100)	Unique SIM serial
iccid	VARCHAR(100)	Unique ICCID
puk_number	VARCHAR(30)	Restricted value
telecom_provider_id	BIGINT	Provider
telecom_account_id	BIGINT	Billing account
telecom_plan_id	BIGINT	Subscription plan
activation_date	DATE	Activation date
contract_start_date	DATE	Contract start
contract_expiry_date	DATE	Contract expiry
monthly_rental	DECIMAL(18,2)	Monthly plan cost
roaming_enabled	BOOLEAN	Roaming control
international_calls_enabled	BOOLEAN	International calls
premium_sms_enabled	BOOLEAN	Premium SMS
current_employee_id	BIGINT	Assigned employee
current_device_id	BIGINT	Assigned device
sim_status	VARCHAR(30)	SIM status

28. Telecom Account Form

28.1 Form Fields

- Telecom Provider
- Company
- Account Name

- Account Number
- Billing Cycle
- Billing Email
- Payment Method
- Credit Limit
- Account Manager
- Contact Number
- Contract Start
- Contract Expiry
- Status

28.2 Database Table

telecom_accounts

- id
- telecom_provider_id
- company_id
- account_name
- account_number
- billing_cycle_day
- billing_email
- payment_method
- credit_limit
- account_manager_name
- account_manager_phone
- contract_start_date
- contract_expiry_date
- status
- remarks

29. Telecom Plan Form

29.1 Form Fields

- Provider
- Plan Name
- Plan Code
- Monthly Charge
- Call Minutes
- Data Allowance
- SMS Allowance
- Roaming Allowance
- Contract Period
- Excess Call Rate

- Excess Data Rate
- Premium SMS Rate
- Cancellation Charge
- Active Status

29.2 Database Table

telecom_plans

- id
 - telecom_provider_id
 - plan_code
 - plan_name
 - monthly_charge
 - included_call_minutes
 - included_data_mb
 - included_sms
 - roaming_allowance
 - contract_months
 - excess_call_rate
 - excess_data_rate
 - premium_sms_rate
 - early_cancellation_charge
 - is_active
-

30. SIM Assignment Form

30.1 Form Fields

- SIM Card
- Mobile Number
- Employee
- Company
- Branch
- Department
- Assignment Date
- Device
- IMEI
- Handover Condition
- Employee Acknowledgement
- Handover Document
- Remarks

30.2 Return Section

- Return Date
- SIM Condition
- Device Condition
- SIM Status after Return
- Cancellation Requested
- Reallocation Required
- Return Remarks

30.3 Database Table

sim_assignments

- id
 - sim_card_id
 - employee_id
 - department_id
 - assigned_device_id
 - assignment_date
 - return_date
 - handover_condition
 - return_condition
 - assignment_status
 - handover_document_id
 - acknowledgement_status
 - assigned_by
 - returned_to
 - remarks
-

31. Telecom Invoice Upload Form

31.1 Form Fields

- Telecom Provider
- Telecom Account
- Invoice Number
- Invoice Month
- Invoice Date
- Due Date
- PDF Invoice
- Excel or CSV File
- Total Before VAT
- VAT
- Invoice Total

- Processing Method
- Processing Status

31.2 Processing Methods

- Automatic PDF extraction
- Excel import
- CSV import
- Manual entry

31.3 Database Table

telecom_invoices

Column	Type	Description
id	BIGINT	Primary key
telecom_account_id	BIGINT	Account
invoice_number	VARCHAR(100)	Invoice reference
invoice_month	DATE	Billing month
invoice_date	DATE	Invoice date
due_date	DATE	Payment due date
amount_before_tax	DECIMAL(18,2)	Before VAT
tax_amount	DECIMAL(18,2)	VAT
total_amount	DECIMAL(18,2)	Total
processing_method	VARCHAR(30)	PDF, Excel or manual
processing_status	VARCHAR(30)	Uploaded, processed or verified
document_id	BIGINT	Invoice file
payment_status	VARCHAR(30)	Paid or unpaid

32. Telecom Invoice Detail Form

32.1 Form Grid Fields

- Mobile Number
- SIM Card
- Employee

- Standard Plan Charge
- Call Charges
- Data Charges
- SMS Charges
- Premium SMS
- Roaming Charges
- International Charges
- Other Charges
- Discount
- VAT
- Total
- Variance
- Variance Reason
- Review Status

32.2 Database Table

telecom_invoice_lines

Column	Type	Description
id	BIGINT	Primary key
telecom_invoice_id	BIGINT	Invoice
sim_card_id	BIGINT	SIM
mobile_number	VARCHAR(30)	Number shown in invoice
employee_id	BIGINT	Assigned employee
plan_charge	DECIMAL(18,2)	Standard plan
call_charge	DECIMAL(18,2)	Call expense
data_charge	DECIMAL(18,2)	Data expense
sms_charge	DECIMAL(18,2)	SMS expense
premium_sms_charge	DECIMAL(18,2)	Premium SMS
roaming_charge	DECIMAL(18,2)	Roaming
international_charge	DECIMAL(18,2)	International calls
other_charge	DECIMAL(18,2)	Other costs
discount_amount	DECIMAL(18,2)	Discount
tax_amount	DECIMAL(18,2)	VAT
total_amount	DECIMAL(18,2)	Total

Column	Type	Description
variance_amount	DECIMAL(18,2)	Actual minus plan
variance_reason	TEXT	Explanation
review_status	VARCHAR(30)	Normal, review or approved

32.3 Alert Rules

An alert should be created when:

- Total exceeds plan by a configured amount.
- Total exceeds plan by a configured percentage.
- Premium SMS charge is greater than zero.
- Roaming charge is greater than zero.
- Unassigned SIM receives charges.
- Cancelled SIM receives charges.
- SIM has zero usage for several months.
- The invoice contains an unknown number.

33. Document Upload Form

33.1 Form Fields

- Related Module
- Related Asset
- Document Type
- Document Number
- Document Name
- Issue Date
- Expiry Date
- File
- Version Number
- Confidentiality Level
- Description
- Active Version
- Remarks

33.2 Database Table

documents

Column	Type	Description
id	BIGINT	Primary key
company_id	BIGINT	Company
module_name	VARCHAR(50)	Module
entity_type	VARCHAR(100)	Vehicle, SIM, employee or asset
entity_id	BIGINT	Related record
document_type_id	BIGINT	Document type
document_number	VARCHAR(100)	Document number
document_name	VARCHAR(200)	Display name
original_file_name	VARCHAR(255)	Original name
stored_file_name	VARCHAR(255)	Stored name
file_path	VARCHAR(500)	Storage location
mime_type	VARCHAR(100)	File type
file_size	BIGINT	File size
issue_date	DATE	Issue date
expiry_date	DATE	Expiry date
version_number	INTEGER	Version
confidentiality_level	VARCHAR(30)	Public, internal or confidential
is_current_version	BOOLEAN	Current version
uploaded_by	BIGINT	User
uploaded_at	TIMESTAMP	Upload time

33.3 Document Security

- Users should only access authorized documents.
- Sensitive documents should be encrypted.
- Deleted files should be archived rather than immediately destroyed.
- Download and print actions should be audited.
- Expired documents should remain available in history.

34. Expense Entry Form

34.1 Form Fields

- Expense Number
- Company
- Branch
- Department
- Cost Centre
- Asset
- Vehicle or SIM
- Expense Category
- Supplier
- Expense Date
- Invoice Number
- Reference Number
- Amount Before VAT
- VAT
- Total Amount
- Payment Status
- Payment Date
- Approval Status
- Invoice Attachment
- Remarks

34.2 Database Table

asset_expenses

- id
- expense_number
- company_id
- branch_id
- department_id
- cost_center_id
- asset_id
- expense_category_id
- supplier_id
- expense_date
- invoice_number
- reference_number
- amount_before_tax
- tax_amount
- total_amount
- payment_status
- payment_date

- approval_status
 - approved_by
 - approved_at
 - document_id
 - remarks
-

35. Alert Configuration Form

35.1 Form Fields

- Alert Name
- Module
- Alert Type
- Related Date Field
- First Alert Before Days
- Second Alert Before Days
- Final Alert Before Days
- Escalation Days
- Email Enabled
- WhatsApp Enabled
- In-System Enabled
- Recipient Roles
- Active Status

35.2 Database Tables

alert_rules

- id
- alert_name
- module_name
- alert_type
- trigger_field
- first_alert_days
- second_alert_days
- final_alert_days
- escalation_days
- email_enabled
- whatsapp_enabled
- system_enabled
- is_active

alerts

- id

- alert_rule_id
- company_id
- entity_type
- entity_id
- alert_title
- alert_message
- due_date
- priority
- alert_status
- generated_at
- acknowledged_by
- acknowledged_at
- completed_by
- completed_at

alert_recipients

- id
 - alert_id
 - user_id
 - employee_id
 - delivery_channel
 - delivery_status
 - delivered_at
 - read_at
-

36. Approval Request Form

36.1 Form Fields

- Request Type
- Related Module
- Related Record
- Requested By
- Request Date
- Amount
- Reason
- Priority
- Supporting Documents
- Current Approval Level
- Status

36.2 Approval Action Fields

- Approver

- Approval Level
- Decision
- Comments
- Decision Date

36.3 Database Tables

approval_requests

- id
- request_number
- company_id
- module_name
- entity_type
- entity_id
- request_type
- requested_by
- requested_at
- request_amount
- reason
- priority
- current_approval_level
- approval_status

approval_actions

- id
- approval_request_id
- approval_level
- approver_user_id
- action
- comments
- action_at

approval_workflows

- id
- workflow_name
- module_name
- request_type
- company_id
- minimum_amount
- maximum_amount
- is_active

approval_workflow_steps

- id

- approval_workflow_id
- step_number
- approver_role_id
- approver_user_id
- is_mandatory
- escalation_hours

37. Audit Trail Structure

37.1 Database Table

audit_logs

Column	Type	Description
id	BIGINT	Primary key
user_id	BIGINT	User
company_id	BIGINT	Company
module_name	VARCHAR(100)	Module
entity_type	VARCHAR(100)	Table or entity
entity_id	BIGINT	Record
action_type	VARCHAR(30)	Create, update, delete or print
previous_values	JSONB	Old values
new_values	JSONB	New values
ip_address	VARCHAR(50)	User IP
user_agent	TEXT	Browser details
action_at	TIMESTAMP	Action time

37.2 Explanation

JSONB should be used for previous and new values because every table has different columns. Audit logs should not be editable through the normal application.

38. Recommended Table Relationships

The principal relationships are:

- One company has many branches.
- One company has many departments.
- One department has many employees.
- One asset category has many asset types.
- One asset has one vehicle record or one SIM record.
- One asset can have many assignments.
- One asset can have many documents.
- One asset can have many expenses.
- One vehicle can have many registration records.
- One vehicle can have many insurance policies.
- One vehicle can have many odometer readings.
- One vehicle can have many maintenance jobs.
- One vehicle can have many fines.
- One vehicle can have many permits.
- One vehicle can have many incidents.
- One vehicle can have many toll transactions.
- One telecom account can contain many SIM cards.
- One SIM can have many assignments.
- One telecom invoice can have many invoice lines.
- One employee can receive multiple assets.
- One supplier can provide services, parts, insurance, or telecom services.

39. Simplified Entity Relationship Diagram

```
companies
|
|---- branches
|
|---- departments
|      |
|      |---- employees
|
|---- assets
|      |
|      |---- asset_assignments
|      |
|      |---- documents
|      |
|      |---- asset_expenses
|
```

```

        |---- vehicles
        |
        |---- vehicle_registrations
        |---- vehicle_insurance_policies
        |---- vehicle_odometer_readings
        |---- vehicle_service_contracts
        |---- vehicle_service_schedules
        |---- maintenance_requests
        |---- maintenance_jobs
        |---- vehicle_parts
        |---- vehicle_tyres
        |---- traffic_fines
        |---- vehicle_permits
        |---- vehicle_incidents
        |---- toll_transactions
    |
    |---- sim_cards
    |
    |---- sim_assignments
    |---- telecom_invoice_lines

telecom_accounts
|
|---- sim_cards
|
|---- telecom_invoices
|
|---- telecom_invoice_lines

toll_accounts
|
|---- toll_tags
|
|---- toll_transactions

```

40. Dashboard Forms and Widgets

40.1 Main Dashboard Filters

- Company
- Branch
- Department
- Asset Category
- Date Range

- Status

40.2 Main Dashboard Cards

- Total Assets
- Assigned Assets
- Unassigned Assets
- Assets Under Maintenance
- Expired Documents
- Expiries Within 30 Days
- Pending Approvals
- Overdue Tasks

40.3 Fleet Dashboard Cards

- Total Vehicles
- Active Vehicles
- Vehicles Under Maintenance
- Registration Expiring
- Insurance Expiring
- Service Due
- Overdue Service
- Pending Fines
- Open Accidents
- Missing Odometer Updates

40.4 Telecom Dashboard Cards

- Total SIM Cards
- Assigned SIM Cards
- Active Unassigned SIM Cards
- Suspended SIM Cards
- Idle SIM Cards
- Plans Above Limit
- Premium SMS Charges
- Roaming Charges
- Contracts Expiring

40.5 Dashboard Lists

- Expiry checklist
- Overdue services
- High telecom usage
- Unassigned active subscriptions
- Latest maintenance requests
- Pending approval requests
- Recent incidents

- Missing documents
-

41. Main System Workflows

41.1 New Vehicle Workflow

1. Create the general asset record.
2. Enter vehicle details.
3. Upload registration and insurance documents.
4. Create the initial registration record.
5. Create the insurance policy record.
6. Configure the service schedule.
7. Assign the vehicle to a driver.
8. Link the Salik or Darb tag.
9. Activate alerts.
10. Mark the vehicle as operational.

41.2 Vehicle Service Workflow

1. Driver or fleet user submits a maintenance request.
2. Fleet manager reviews the request.
3. Estimate is obtained.
4. Approval request is generated.
5. Authorized user approves or rejects.
6. Maintenance job is opened.
7. Parts and labour are recorded.
8. Invoice and job card are uploaded.
9. Odometer and service schedule are updated.
10. Expense entry is generated.
11. Vehicle status returns to active.
12. Job is closed.

41.3 Vehicle Renewal Workflow

1. Expiry alert is generated.
2. Renewal task is assigned.
3. Insurance quotation is obtained.
4. Vehicle passing is completed.
5. Renewal costs are entered.
6. Approval is completed.
7. New registration document is uploaded.
8. New expiry date is recorded.
9. Previous registration remains in history.
10. Renewal task is closed.

41.4 Traffic Fine Workflow

1. Fine is entered.
2. Vehicle and driver are identified.
3. Payment responsibility is assigned.
4. Manager reviews the fine.
5. Fine is paid by the company or employee.
6. Salary deduction is recorded when applicable.
7. Receipt is uploaded.
8. Fine status changes to cleared.

41.5 New SIM Workflow

1. Create a general asset record.
2. Enter SIM details.
3. Select telecom account and plan.
4. Upload agreement.
5. Assign the SIM to an employee.
6. Record the device and IMEI.
7. Generate a handover form.
8. Obtain employee acknowledgement.
9. Mark the SIM as active and assigned.

41.6 Telecom Invoice Workflow

1. Upload monthly invoice.
2. Extract or import invoice details.
3. Match mobile numbers with SIM records.
4. Match SIM cards with employees.
5. Calculate plan-versus-actual variance.
6. Flag premium SMS, roaming, and excess charges.
7. Accounts team reviews the records.
8. Department manager explains unusual charges.
9. Invoice is approved for payment.
10. Expense records are created.
11. Management reports are updated.

42. API Structure Recommendation

Example REST API endpoints:

Asset APIs

```
GET    /api/assets
GET    /api/assets/:id
POST   /api/assets
PUT    /api/assets/:id
DELETE /api/assets/:id
POST   /api/assets/:id/assign
POST   /api/assets/:id/return
GET    /api/assets/:id/history
```

Vehicle APIs

```
GET    /api/vehicles
GET    /api/vehicles/:id
POST   /api/vehicles
PUT    /api/vehicles/:id
GET    /api/vehicles/:id/registrations
POST   /api/vehicles/:id/registrations
GET    /api/vehicles/:id/maintenance
POST   /api/vehicles/:id/odometer-readings
POST   /api/vehicles/:id/fines
POST   /api/vehicles/:id/incidents
```

Telecom APIs

```
GET    /api/sim-cards
POST   /api/sim-cards
PUT    /api/sim-cards/:id
POST   /api/sim-cards/:id/assign
POST   /api/sim-cards/:id/return
GET    /api/telecom-invoices
POST   /api/telecom-invoices/upload
POST   /api/telecom-invoices/:id/process
GET    /api/telecom-invoices/:id/exceptions
```

Document APIs

```
POST   /api/documents/upload
GET    /api/documents/:id
GET    /api/documents/:id/download
```

```
GET    /api/documents/:id/print
DELETE /api/documents/:id
```

Dashboard APIs

```
GET /api/dashboard/summary
GET /api/dashboard/fleet
GET /api/dashboard/telecom
GET /api/dashboard/expiries
GET /api/dashboard/alerts
GET /api/dashboard/pending-approvals
```

43. Database Constraints and Indexes

43.1 Unique Constraints

Unique constraints should be applied to:

- Company Code
- Branch Code within the same company
- Employee Code
- Asset Code
- Chassis Number
- Plate Number plus Plate Emirate
- Toll Tag Number
- SIM Serial Number
- ICCID
- Mobile Number
- Telecom Account Number
- Fine Reference Number
- Registration Number where appropriate
- Invoice Number within the same supplier or provider

43.2 Recommended Indexes

Indexes should be created for:

- asset_code
- asset_status_id
- company_id
- branch_id
- department_id

- current_employee_id
- plate_number
- chassis_number
- registration expiry date
- insurance expiry date
- permit expiry date
- service due date
- next service odometer
- mobile_number
- sim_status
- contract expiry date
- invoice month
- employee_id
- document expiry date
- alert status
- approval status

43.3 Financial Data Types

Use:

```
DECIMAL(18,2)
```

for monetary amounts.

Do not use floating-point types for financial values.

43.4 Date and Time Types

Use:

- DATE for business dates
- TIMESTAMP WITH TIME ZONE for system activity
- TIME for transaction times

44. Soft Delete and Record Retention

Important records should not be physically deleted.

Tables should include:

- deleted_at
- deleted_by

- deletion_reason

Soft deletion should apply to:

- Assets
- Vehicles
- SIM cards
- Employees
- Assignments
- Documents
- Expenses
- Maintenance jobs
- Fines
- Incidents
- Invoices

Financial records, approvals, audit logs, and historical assignments should normally never be permanently deleted.

45. Security Requirements

The system should provide:

- Role-based access control
- Company-level data restrictions
- Branch-level data restrictions
- Field-level protection for confidential data
- Password hashing
- Session expiration
- Login attempt controls
- File access authorization
- Secure document URLs
- Audit logging
- Backup encryption
- HTTPS
- Database connection encryption

Sensitive fields include:

- Emirates ID
 - Passport details
 - PUK numbers
 - Employee personal information
 - Supplier banking information
 - Telecom account credentials
-

46. Backup and Recovery

The system should support:

- Daily database backup
- Weekly full backup
- Document storage backup
- Off-site backup
- Backup encryption
- Backup retention policy
- Restore testing
- Disaster recovery procedure

Recommended retention:

- Daily backups: 30 days
 - Weekly backups: 12 weeks
 - Monthly backups: 12 months
 - Annual backups: based on company policy
-

47. Recommended Development Phases

Phase 1: Foundation

- Company
- Branch
- Department
- Employee
- User
- Roles and permissions
- Asset categories
- General asset register
- Document management

Phase 2: Vehicle Management

- Vehicle master
- Driver assignment
- Registration
- Insurance
- Odometer
- Service schedule
- Maintenance
- Fines

- Permits
- Incidents

Phase 3: Toll Management

- Toll accounts
- Toll tags
- Salik import
- Darb manual entry
- Toll reports

Phase 4: Telecom Management

- Telecom accounts
- Telecom plans
- SIM register
- SIM assignments
- Invoice upload
- Invoice detail processing
- Cost-control alerts

Phase 5: Workflow and Reporting

- Approval workflows
- Alert engine
- Email and WhatsApp notifications
- Dashboards
- Management reports
- Audit reports
- One-click printing

48. Final Functional Result

Once implemented, the Asset Management System will allow management to answer the following questions immediately:

- What assets are currently owned or managed?
- Which company and department control each asset?
- Which employee is responsible for the asset?
- Which vehicles require registration renewal?
- Which insurance policies are expiring?
- Which vehicles are due for service?
- Which vehicles have pending fines or accidents?
- What is the total cost of each vehicle?
- Which Salik or Darb account is linked to each vehicle?

- Which SIM belongs to each employee?
- Which active SIM cards are unassigned?
- Which telecom lines are exceeding their plans?
- Which documents are missing or expired?
- Which approvals are pending?
- Which tasks are overdue?
- What is the complete history of each asset?

All vehicle, fleet, toll, telecom, SIM, maintenance, expense, document, assignment, renewal, compliance, approval, and audit information should be maintained under this centralized Asset Management System.